

Aejandro De La Torre

LIS 440

Lecture: Week 14 | Pandas Aggregation, Data Cleaning, and Intro to Visualization

12/2/25

Code

```
# Import the necessary packages
import pandas as pd
import os
from pathlib import Path

# Print version of pandas
print("Alejandro De La Torre's code....")
print(pd.__version__)

# -----
# File & Path Setup
# -----

# Set root directory (two levels up from this script)
root = Path(__file__).parent.parent.resolve()

# Show current working directory
print("Current Working Directory:", Path.cwd())

# Define key directories
data = root / 'data'
data_raw = data / 'raw'
data_processed = data / 'processed'
output = root / 'output'
code = root / 'code'

# -----
# Grouping Tutorial: Ice Cream Sales
# -----

# Fetch the data
icecreamDF = pd.read_csv(data_processed / 'icecreamsales25.csv')
print("\nIce Cream Data: ")
print(icecreamDF)

# Fetch the Flavor Column
flavorColumn = icecreamDF["Flavor"]
print("Flavor Column:")
print(flavorColumn)

# Get just the chocolate of the flavor column
print()
chocolateRows = icecreamDF[ flavorColumn == "Chocolate" ]
print(chocolateRows)
```

```
# Now to sort by flavor:
print()
print("Sort and group by flavor")
flavorGrouping = icecreamDF.groupby("Flavor")
flavorGroupingTotals = flavorGrouping["Sale"].sum()
print(flavorGroupingTotals)
```

Shell

```
>>> %Run icecream_sales_grouping.py
Alejandro De La Torre's code....
2.3.3
Current Working Directory: /Users/alexdelatorre/Library/CloudStorage/OneDrive-UW-Madison/Coursework/L I S
440/Lectures/Week 14/Tuesday/code
```

Ice Cream Data:

	Transaction	Flavor	Container	Sale
0	1	Strawberry	SugarCone	1
1	2	Vanilla	Cup	1
2	3	Blue Moon	Cup	3
3	4	Strawberry	Cup	1
4	5	Chocolate	SugarCone	2
5	6	Strawberry	Cup	1
6	7	Strawberry	SugarCone	3
7	8	Strawberry	WaffleCone	2
8	9	Strawberry	Cup	3
9	10	Vanilla	Cup	3
10	11	Chocolate	Cup	2
11	12	Chocolate	SugarCone	3
12	13	Chocolate	WaffleCone	2
13	14	Chocolate	WaffleCone	2
14	15	Vanilla	SugarCone	1
15	16	Chocolate	WaffleCone	2
16	17	Vanilla	SugarCone	3
17	18	Blue Moon	SugarCone	3
18	19	Blue Moon	SugarCone	1
19	20	Blue Moon	Cup	1
20	21	Blue Moon	WaffleCone	1
21	22	Blue Moon	SugarCone	3
22	23	Strawberry	SugarCone	3
23	24	Vanilla	WaffleCone	3
24	25	Blue Moon	WaffleCone	1

Flavor Column:

```
0 Strawberry
1 Vanilla
2 Blue Moon
3 Strawberry
```

```
4 Chocolate
5 Strawberry
6 Strawberry
7 Strawberry
8 Strawberry
9 Vanilla
10 Chocolate
11 Chocolate
12 Chocolate
13 Chocolate
14 Vanilla
15 Chocolate
16 Vanilla
17 Blue Moon
18 Blue Moon
19 Blue Moon
20 Blue Moon
21 Blue Moon
22 Strawberry
23 Vanilla
24 Blue Moon
```

```
Name: Flavor, dtype: object
```

```
Transaction  Flavor  Container  Sale
4           5  Chocolate  SugarCone    2
10          11  Chocolate    Cup      2
11          12  Chocolate  SugarCone    3
12          13  Chocolate  WaffleCone    2
13          14  Chocolate  WaffleCone    2
15          16  Chocolate  WaffleCone    2
```

```
Sort and group by flavor
```

```
Flavor
```

```
Blue Moon    13
```

```
Chocolate    13
```

```
Strawberry   14
```

```
Vanilla      11
```

```
Name: Sale, dtype: int64
```

```
>>>
```